CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fifth Semester of B.Tech Examination (IT)

November 2012

IT302: Design & Analysis of Algorithm

Date: 28.11.2012,Wednesday **Time: 10:00 a.m to 1:00 p.m** **Maximum Marks:70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION-I

Q-1

(a) Which version of algorithm is preferred when memory resources are limited, *iterative* or *recursive*? Justify your answer. **2**

(b) Sort following in increasing order of running time for large values of n: **2**

$n^3, 2^n, \log_2 n, n^2 \log n, n^n, n!-(n-1)!$

(c) Mention the parameters for finding suitable algorithm among many candidate algorithms. Justify parameter with suitable example. **3**

Q-2

(a) Explain the following functions with respect to Greedy Algorithms: **4**
1. Feasibility function, 2. Selection function 3. Solution function 4. Objective function.

(b) Find out No. of Primitive Operations and Growth Rate of Following Algorithm for Best, Worst and Average Case. **3**

Algorithm:

```
found = 0;
I = 0;
while (!found && i < N)
{  If (key ==X[i]) found = 1;
   i++;
}
```

(c) The array given as input to a quick sort algorithm is {1, 2, 3, 4, 5, 6, 7, 8, 9, 10} What will be the running time of quick sort if the pivot is taken to be (i) the first element and (ii) median of three (first, middle, last) elements ? Write recurrence for above two cases and find its time complexity. **7**

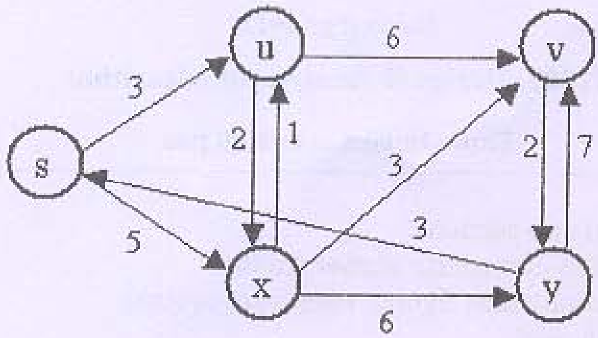
OR

(c) Solve using Recurrence Tree Method. **7**
T (n) =4T (n/3) +n, where n is in multiple of 3.
T (n) =3T (n/2) +n, where n is in multiple of 2.

Q-3 Attempt Any Two. **14**

(a) Given a set S of n activities and start time, S_i and finish time F_i of an i^{th} activity. Write an algorithm to find the maximum size set of mutually compatible activities using Greedy approach. Also find time Complexity of the algorithm. Let 11 activities are given S = {p, q, r, s, t, u, v, w, x, y, z} start and finished times for proposed activities are (1, 4), (3, 5), (0, 6), (5, 7), (3, 8), (5, 9), (6, 10), (8, 11), (8, 12),(2, 13) and (12, 14).
Find the maximum size set of mutually compatible activities using Greedy algorithm.

(b) Execute the dijsktra algorithm on the following graph, with single-source S.



(c) Solve the following Recurrence equations:

1. $F_n = F_{n-1} + F_{n-2}$, $F_1 = F_2 = 1$
2. $t_n = 2t_{n-1} + 1$, $t_0 = 1$.

SECTION-II

Q-4

(a) Define the Following Terms: 2

(i) Tree Edge (ii) Back Edge

(b) An asymptotically fast algorithm running on Slow computer is better than asymptotically slow algorithm is running on Fast computer. (T/F), Justify your answer. 2

(c) If $f(n) = 3n^3 + 5n + 5$ and $g(n) = 5n^2 + 3n + 5$ then check following equalities: 3

1. $f(n) = \theta(g(n))$ 2. $g(n) = \Omega(f(n))$ 3. $g(n) = O(f(n))$

Q-5

(a) Write an algorithm for matrix chain multiplication. 4

(b) Explain String matching problem. Write its applications. 3

(c) Generate at least 2-solution for 5-Queen's problem. Draw a pruned state space tree for the problem. 7

OR

(c) Solve using Large integer multiplication method : 4568 X 3214 7

Q-6 Attempt Any Two. 14

(a) A company has 4 machines available for assignment to 4 tasks. Any machine can be assigned to any task, and each task requires processing by one machine. The time required to set up each machine for the processing of each task is given in the table below.

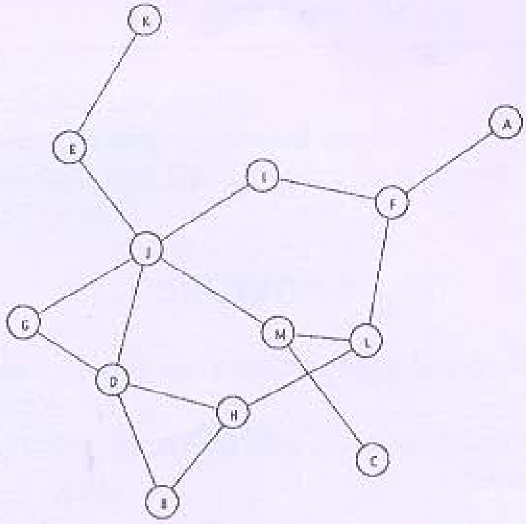
TIME (Hours)

	Task 1	Task 2	Task 3	Task 4
Machine 1	13	4	7	6
Machine 2	1	11	5	4
Machine 3	6	7	2	8
Machine 4	1	3	5	9

The company wants to minimize the total setup time needed for the processing of all four tasks. Find the solution using branch & bound method.

(b) Define Longest common subsequence problem. Find the length of the subsequence for X (0 1 1 1 0 1 1) and Y (1 1 1 1 0 0 1). Also write algorithms for finding longest common subsequence and length of the longest common subsequence.

(c) Write an algorithm for Depth First Search on undirected graph. Run The Depth First Search algorithm on the following graph from source vertex H.



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